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THE LITHIUM-DEPLETION BOUNDARY AND THE AGE OF THE YOUNG OPEN CLUSTER IC 2391

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ABSTRACT

In previous programs, we have identified a large number of possible, very low mass members of the young open cluster IC 2391 based on their location in an I versus $(R-I)_c$ color-magnitude diagram. We have now obtained new photometry and intermediate-resolution ($\Delta\lambda = 2.7$ Å) spectra of 19 of these objects ($14.9 \leq I_c \leq 17.5$) in order to confirm cluster membership. We identify 15 of our targets as likely cluster members based on their *VRI* photometry, spectral types, radial velocity, and Hα emission strengths. Higher signal-to-noise ratio spectra were obtained for eight of these probable cluster members in order to measure the strength of the doublet Li λ6708 and thus obtain an estimate of the cluster's age. One of these eight stars has a definite lithium detection, and two other (fainter) stars have possible lithium detections. A color-magnitude diagram for our program objects shows that the lithium-depletion boundary in IC 2391 is at $I_c = 16.2$. Using recent theoretical model predictions, we derive an age for IC 2391 of 53 ± 5 Myr. While this is considerably older than the age most commonly attributed to this cluster (~35 Myr), this result for IC 2391 is comparable to those recently derived for the Pleiades and α Persei clusters and can be explained by new models for high-mass stars that incorporate a modest amount of convective-core overshooting.

Subject headings: open clusters and associations: individual (IC 2391) — stars: low-mass, brown dwarfs

1. INTRODUCTION

Measurement of the location of the "lithium-depletion boundary" (LDB) offers a new method for determining the age of young open clusters that is supposedly more accurate and less subject to possible systematic errors than any of the other methods (cf. Bildsten et al. 1997; Basri 1997; Ventura et al. 1998a, 1998b; Ushomirsky et al. 1998). The LDB method was first applied to the Pleiades cluster (Basri, Marcy, & Graham 1996; Rebolo et al. 1996), with the most recent determination (Stauffer, Schultz, & Kirkpatrick 1998) yielding an age of 125 ± 8 Myr. That age is considerably older than some estimates of the Pleiades age but is consistent with other estimates that assume a modest amount of convective-core overshoot (Ventura et al. 1998a, 1998b). Our group has now obtained LDB ages for two other open clusters: α Persei (Stauffer et al. 1999; see also Basri & Martín 1999) and IC 2391. We report the results for IC 2391 in this Letter.

Located at a distance of about ~155 pc, IC 2391 ($\alpha = 8^h40^m.2$, $\delta = -53°04'$, J2000.0) is the fifth nearest cluster to the Sun. The first faint proper-motion survey of this cluster was obtained by Stauffer et al. (1989), who yielded nine previously undiscovered late-type members. Patten & Simon (1996) used data from *ROSAT* to obtain a more complete membership list by combining the X-ray data with ground-based optical photometry and spectroscopy. Additional spectroscopic studies by Stauffer et al. (1997) confirmed that the cluster is relatively young based on the rotational velocities, lithium abundances, and chromospheric activity for G, K, and early-M dwarfs. However, none of those studies provided cluster members that were faint enough ($I > 15$) to be suitable for the LDB test. Recently, Patten & Pavlovsky (1999) and Barrado y Navascués et al. (1999a) have provided lists of faint candidate members of IC 2391 that may extend to a low enough mass. Spectra of some of these candidates are used here to determine the location of the LDB in IC 2391.

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2. OBSERVATIONS AND DATA REDUCTION

Spectroscopic observations of the most promising candidate low-mass members of the cluster were carried out with the 4 m telescope at CTIO during 1999 January 9–11. We used the Ritchey-Chrétien spectrograph with the Blue Air Schmidt camera, the KPGLD grating (790 line mm^{-1}), the Loral 3k CCD, and a 1″ slit. This configuration yielded a spectral resolution of 2.7 Å and a free spectral range of 6295–8815 Å.

The primary targets for our observations were selected from the optical survey by Barrado y Navascués et al. (1999a), which identified several dozen very low mass stars and brown dwarf candidates based primarily on R_c and I_c photometry. The targets were selected for having apparent magnitudes near to or below the expected location of the LDB in IC 2391, using the nominal age and distance to the cluster and the theoretical model predictions of Chabrier & Baraffe (1997). Additional *V*-filter data, obtained at the CTIO 0.9 m telescope using the Tektronix 2k No. 3 camera during 1999 January 4–6, allowed us to refine the target list based on their location in a $[(V-R_c), (R-I)_c]$ color-color diagram. For some of our objects, near-infrared data from the 2MASS project (J. Adams 1998, private communication) were also used to refine the target list. Additional details can be found in Barrado y Navascués et al. (1999a) and Barrado y Navascués, Stauffer, & Briceño (1999c). Several presumably higher mass objects were added to our target list from the surveys of Patten & Pavlovsky (1999) and Patten & Simon (1996). We also observed several Gliese stars of spectral types M0–M6 and BRI 0021−0214, a M9.5 spectral type dwarf (Basri & Marcy 1995), in order to serve as template objects with which to compare the IC 2391 spectra. For the IC 2391

TABLE 1
DATA FOR IC 2391 VERY LOW MASS STARS AND BROWN DWARF CANDIDATES

NAME[a]	V	I_c	$(R-I)_c$		EW (Å)		RADIAL VELOCITY (km s^{-1})		SPECTRAL TYPE	Li I	MEMBER
			Photometry	Spectroscopy	Na I	Hα	Hα	Cross-Correlation[b]			
VXR 27	13.97	11.51	1.33	1.30	...	<0	...	−1[c]	M2.5	No	No
VXR 71a	15.29	12.90	1.33	1.40	...	4.9	−25	−9	M3	No	Yes (SB?)
CTIO-152	18.11	14.89	1.78	1.65	5.1	9.9	31	25:	M4	No	Yes
PP 11	18.48	15.30	1.79	1.75	4.7	5.4	19	25:	M4.5	No	Yes
CTIO-094	18.27	15.31	1.74	1.50	4.9	<0	...	24:	M3.5	...	No
PP 14	19.22	15.83	1.87	1.85	4.8	6.3	18	17:	M5	No	Yes
CTIO-136[d]	19.72	15.87	1.98	1.85	5.2	6.1	31	27:	M5.5[e]	...	Yes
CTIO-017	19.30	15.99	1.75	1.70	5.6	11.0	4	9:	M4[e]	...	Yes
CTIO-059[f]	19.47	16.05	1.82	1.70	5.6	49.5/18.8	106/42	18/25:	M4	No	Yes
CTIO-096[g]	19.75	16.14	1.91	1.90	5.3	8.7	31	22:	M5	No	Yes
CTIO-205	20.21	16.21	2.07	2.00	5.2	8.0	−14	8	M5.5[e]	...	Yes?
CTIO-038	19.90	16.29	1.91	1.90	5.1	7.0	15	14:	M5	Yes	Yes
CTIO-077	20.04	16.31	1.93	1.85	5.1	7.0	48	19:	M5	...	Yes
CTIO-081	20.38	16.47	2.08	2.05	5.8	8.6	3	2	M5.5	...	Possible
CTIO-041	20.46	16.55	1.92	1.95	5.2	8.7	34	15:	M5	Yes	Yes
CTIO-106	...	16.56	2.01	1.90	5.8	14.1	3	−1:	M5.5	...	Possible
CTIO-062[f]	20.84	16.75	2.05	1.95	5.3	6.6/7.9	10/7	13/11:	M6[e]	Yes	Yes
CTIO-145	...	16.94	2.14	2.15	5.5	7.9	11	13	M6	...	Yes
CTIO-113	21.88	17.28	2.14	...	6.3	4.6	11	−1:	M7[e]	...	Yes
CTIO-061	...	17.31	2.14	2.10	5.9	2.3	46	24	M6[e]	...	Yes
CTIO-040	21.65	17.46	2.12	1.95	6.2	<0	...	0:	M6[e]	...	No

[a] VXR: Patten & Simon 1996; CTIO: Barrado y Navascués et al. 1999a; and PP: Patten & Pavlovsky 1999.
[b] A colon designates an estimated error greater than 25 km s^{-1}.
[c] From Stauffer et al. 1997.
[d] Also known as PP 17.
[e] Uncertainty of 1 subclass in the derived spectral type.
[f] EW(Hα) and radial velocity for two consecutive nights.
[g] Also known as VXRP32h.

low-mass candidates, the exposure times ranged from 10 minutes to 4.5 hr. The data were reduced using standard routines within the IRAF.[2]

3. ANALYSIS

For our program stars, we have determined equivalent widths and line profiles for several spectral features, namely, Hα in emission, Li I λ6708, and the K I and Na I doublets ($\lambda\lambda$7665, 7699 and $\lambda\lambda$8183, 8195, respectively). In addition, we have measured several spectral indices, namely, PC1, PC2, and PC3 at 6525–7050, 7030–7580, and 7540–8265 Å, respectively (Martín, Rebolo, & Zapatero-Osorio 1996); VO at 7350–7560 Å (Kirkpatrick, Henry, & Simons 1995); and TiO and CaH at 6510–7050 Å (Martín & Kun 1996), in order to estimate spectral types and $(R-I)_c$ colors. These estimates were achieved by a calibration that is based on a group of cool dwarfs of known spectral types and colors. The derived spectral types are estimated to have an uncertainty of 0.5 or 1 subclass. Radial velocity estimates were derived for the program stars by two methods: (1) by cross-correlation versus the spectra of GJ 285 and GJ 406 and (2) by measurement of the wavelength of the Hα emission line. The radial velocity errors for the cross-correlation method are estimated to range from 10 to 50 km s^{-1}. The uncertainty in the radial velocity derived from the fit to the Hα profile should be $\sim$15 km s^{-1} for most of our objects based on the repeatability in the measured position of OH night-sky emission lines of similar strength in the sky spectra. Table 1 lists the names of our targets along with the photometry (Barrado y Navascués et al. 1999a) and the spectroscopic results.

[2] IRAF is distributed by National Optical Astronomy Observatories, which is operated by the Association of Universities for Research in Astronomy, Inc., under contract to the National Science Foundation.

3.1. *Membership Criteria*

In the last column of Table 1, we provide an estimate of the membership status for each of our target stars. This status was determined by consideration of the following criteria:

1. A comparison of the photometrically measured $(R-I)_c$ color and the color estimated from the spectral type, assuming $E(R-I)_c \sim 0.01$ for IC 2391 (e.g., Hogg 1960; Patten & Simon 1996), was used to eliminate background objects.
2. Radial velocities were compared with the cluster average of $\sim$15 km s^{-1} (Stauffer et al. 1997).
3. Hα equivalent widths were compared with typical values for field stars and Pleiades members.
4. The strength of the gravity-dependent doublet Na I λ8200, which is strong in dM stars, weaker in pre–main-sequence (PMS) M stars, and very weak in M-type giants (Martín et al. 1996), was used to eliminate background giants and nearby field stars (see Fig. 1).

These criteria allowed us to reject two stars as members of the cluster and to classify another two stars as possible nonmembers. CTIO-040 and CTIO-094 have radial velocities of less than 3 km s^{-1}. For these two stars, the differences in their photometrically measured $(R-I)_c$ color and the color estimated from the spectral type are 0.17 in the case of CTIO-040 and 0.24 for CTIO-094. Both of these stars also have Hα in absorption. In the spectral type range we have covered, all members of the Pleiades have Hα in emission with EW $\sim$ 2–10 Å (Stauffer, Liebert, & Giampapa 1995; Jones, Shetrone, & Fischer 1996). Because IC 2391 is younger, we expect to see similar or stronger Hα emission from our candidates. Therefore, we consider CTIO-040 and CTIO-094 to be nearly certain nonmembers. The targets CTIO-081 and CTIO-106 were found to have radial velocities of less than 3 km s^{-1}. These stars are considered possible nonmembers. The remaining 15 targets in

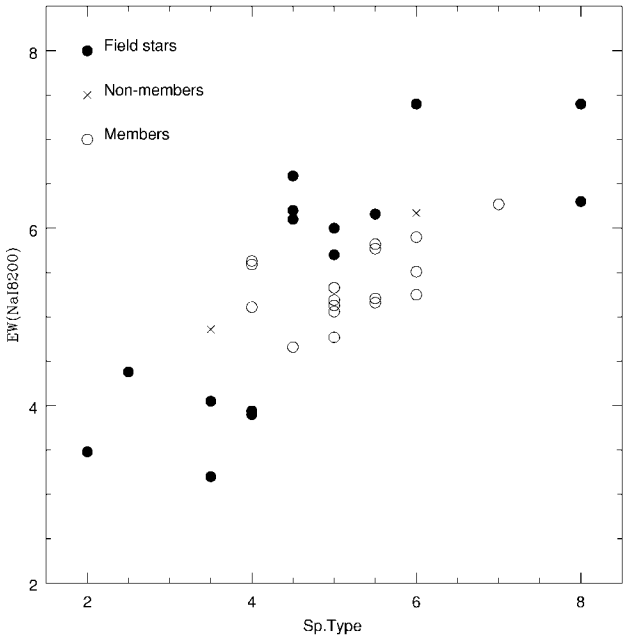


FIG. 1.—Na I λ8200 equivalent width vs. spectral type. The field dwarfs are shown as filled circles, and the IC 2391 members are shown as open circles. The stars that were initially identified as candidate IC 2391 members but that subsequently were rejected as members using the membership criteria outlined in § 3.1 appear as crosses.

our sample appear to be young PMS objects and thus are likely members of the IC 2391 cluster.

3.2. *The Lithium-Depletion Boundary in IC 2391*

Figure 2 shows the region around the lithium doublet for three members whose *I* magnitudes place them near the LDB: PP 14, CTIO-096, and CTIO-038. The spectrum of GJ 406, an M6 V star without lithium, is shown for comparison purposes. Clearly, CTIO-038 ($I_c = 16.29$) contains lithium, whereas PP 14, which is 0.46 mag brighter, does not have a discernible feature at 6708 Å. CTIO-096, which is slightly brighter than CTIO-038, has also apparently depleted all of its lithium. We also have low signal-to-noise ratios (S/Ns) and possible detections of lithium in two other candidate members of IC 2391 (CTIO-041 and CTIO-062), both of which are fainter and redder than CTIO-038. Other candidate cluster members, which are brighter and warmer than CTIO-038 and observed at high S/Ns, do not show lithium in their spectra (see Table 1). Figure 3 shows an $I_c/(R-I)_c$ color-magnitude diagram for our sample of candidates. Based on the detection of lithium in CTIO-038 and the nondetection of lithium in CTIO-096, we can estimate the location of the LDB in IC 2391 to be at $I_c = 16.2 \pm 0.15$, $(R-I)_c = 1.91$.

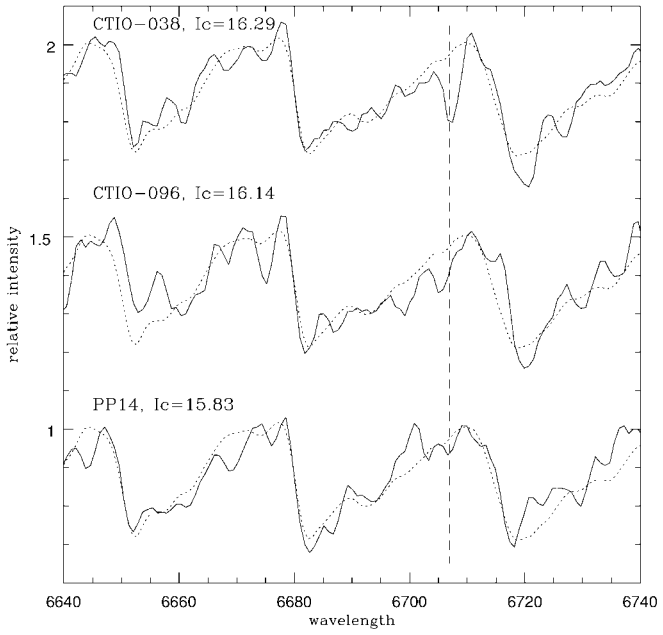


FIG. 2.—Spectra of IC 2391 candidates (*solid lines*). We only show here the spectral region around the doublet Li I λ6708 (*vertical dashed line*). For comparison, GJ 406 (the field M6 dwarf with no lithium) is also plotted (*dotted line*).

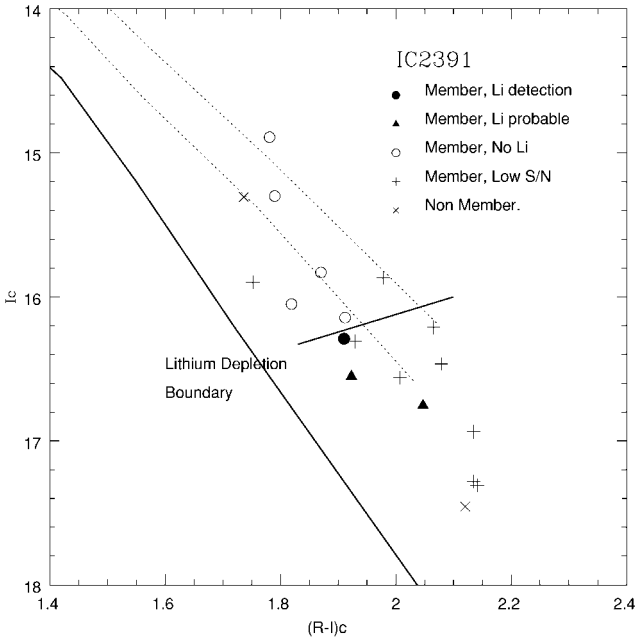


FIG. 3.—Color-magnitude diagram of IC 2391. The probable members observed at low S/Ns appear as plus signs, whereas the members observed at high S/Ns are displayed as circles (filled if lithium was detected, open if not). The filled triangles indicate probable lithium detections. Nonmembers appear as crosses. The location of the LDB is indicated as well as a zero-age main-sequence star (*solid line*) and 50 and 30 isochrones (*dotted lines*; D'Antona & Mazzitelli 1994).

Using the evolutionary models of I. Baraffe (1998, private communication), we can estimate the age of IC 2391 by using the position of the LDB. Following Figure 3 of Stauffer, Schultz, & Kirkpatrick (1998), we have converted the I_c magnitude of the LDB into an age, assuming an interstellar absorption of $A_I = 0.02$ and a distance modulus of $(m-M)_0 = 5.95 \pm 0.1$[3] yielding $M_I = 10.25$, which implies an age of 53 ± 5 Myr. While this value of the cluster's age is significantly larger than previous estimates (~35 Myr), this correction factor to the classical age is consistent with those estimated for other young open clusters, such as the Pleiades (Stauffer et al. 1998) and α Persei (Stauffer et al. 1999), using the LDB method.

[3] This value for the distance of IC 2391 represents a compromise between various determinations of the distance modulus and *Hipparcos* parallaxes (see, e.g., Becker & Fenkart 1971, Lynga 1987, Patten & Pavlovsky 1999, and Robichon et al. 1999 for additional details).

3.3. *The Lowest Mass Members of IC 2391*

At an age of 53 Myr, a star at the LDB should have a mass of ~0.12 $M_\odot$ (Chabrier & Baraffe 1997). The corresponding I_c magnitude for a 0.075 $M_\odot$ object (i.e., an object at the transition boundary between the stellar and substellar domains) in IC 2391 is then $M_I \sim 11.15$. If the latter estimate is correct, then the faintest two candidate members in IC 2391 that we have observed spectroscopically (CTIO-061 and CTIO-113) would be brown dwarfs.

4. DISCUSSION

The ages for young open clusters have been controversial for many years because of the uncertainties in the amount of convective-core overshoot to include in models of high-mass stars. Although an age scale for clusters using models without convective-core overshoot has generally been adopted (Patenaude 1978; Mermilliod 1981), models that incorporate a relatively large convective-core overshoot parameter, yielding cluster ages up to twice as old as the nonovershoot models, have also been advocated (Mazzei & Pigatto 1988). Without additional observational data, it has been difficult to choose between these age scales. The lithium-depletion age method provides the means to resolve this debate. We now have lithium-depletion ages for three open clusters: the Pleiades (Stauffer et al. 1998), α Persei (Basri & Martín 1999; Stauffer et al. 1999), and IC 2391 (this Letter). When compared with the nonovershoot model ages of Mermilliod (1981), the ratios of the LDB age determination to the nonovershoot model age are 1.61, 1.65, and 1.46 for Pleiades, α Persei, and IC 2391, respectively. This suggests that (1) some convective-core overshoot is needed in evolutionary models for high-mass stars and that (2) the amount of convective-core overshoot is not a strong function of mass (at least in the mass range sampled at the turnoff of these three clusters). A revised age scale for open clusters (see Barrado y Navascués, Stauffer, & Bouvier 1999b) would have important implications for a variety of stellar evolution topics including, for example, the empirical (and theoretical) correlations between lithium abundance, coronal activity, and rotation rate as a function of age for low-mass stars.

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